Grounded Study Item Generation from Personal Notes

An Obsidian Plugin with a Multi-Agent Pipeline & Per-Concept Evaluation

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Code, paper, slides
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Problem

Personal note vaults (Obsidian, Notion, Logseq) accumulate concept notes, PDFs, and figures, but turning them into study items — problems, flashcards, code challenges — is manual, time-consuming, and error-prone.

LLMs draft items cheaply but routinely drift from the exact content the learner stored, breaking instructional trust. Grading drift is worse: a single scalar score punishes every concept on a failed answer, including ones the student never even tried to use.

Goal

Generate study items grounded in the learner's own notes, grade them at the concept level, and never move raw notes outside the vault.

Approach

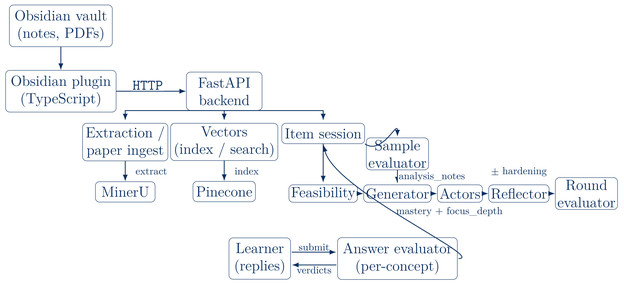
The vault is the source of truth. A TypeScript plugin sends curated JSON views of selected concepts to a FastAPI service that runs a multi-agent loop under tool-calling (Anthropic Claude Haiku 4.5). All outputs are written back to the same vault as markdown.

- Sample Evaluator pre-grades a learner-supplied reference — markdown or an image of an exam clip — and threads pedagogical `analysis_notes` into the Generator.
- Core hardening loop (Generator → Actor → Reflector) refines items until they survive simulated solving; a back-edge re-emits when difficulty ≥ hard.
- Answer Evaluator returns per-concept verdicts: `correct`, `alt_path`, `misunderstood`, `untested`.
- Mastery + depth-aware scheduler: per-concept Elo updates; mastered concepts graduate to integration items; `focus_depth` advances along a topological ordering produced by a link-architect over MinerU-extracted PDF sections.

Why per-concept verdicts

Most LLM evaluators emit a scalar. We emit a list of `ConceptVerdict` objects so an alternate solution path still credits the concept it exercised, and a wrong answer never penalizes concepts the learner did not touch — directly feeding the Elo update and the difficulty progression.

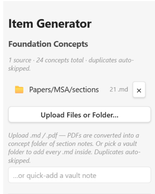
System Architecture



Vault → plugin → FastAPI; multimodal Sample Evaluator pre-grades references into ***analysis_notes***; the staged generation loop hardens items; per-concept Answer Evaluator verdicts feed mastery + depth-aware scheduling back into the next round.

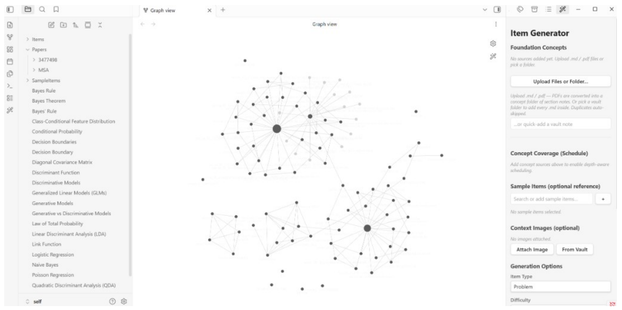
Agent roles

Feasibility	Routes a request to `abandon` / `review` / `generate`.
Generator	Tool-calling JSON; outputs validated against declared concepts.
Actors×3	Simulate solving to surface failure modes.
Reflector	Quality score + bounded refinement; hardening directives.
Round eval	Per-round summary; supports difficulty progression.
Sample eval	Pre-grades md/image reference → `analysis_notes`.
Answer eval	Per-concept verdicts → Elo mastery.
Link-architect	Topological depth over PDF sections (ingest).

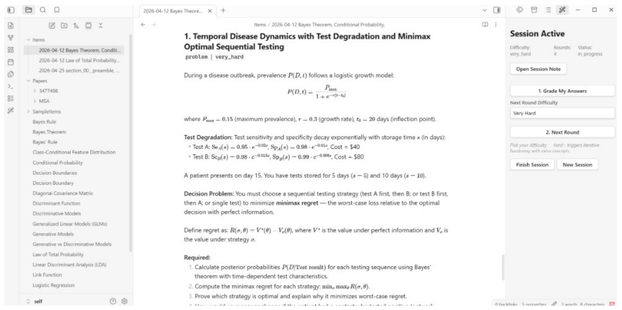


One paper PDF → 21 section concepts, deduplicated.

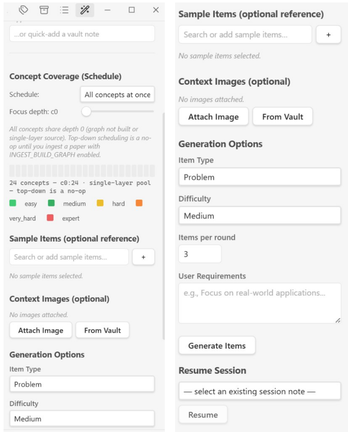
Plugin in Action



Obsidian global graph view — each concept becomes a node; the Item Generator panel reads selected sources directly from the vault.



A generated very_hard problem combining Bayes' theorem with sequential testing under time-degrading sensitivity. LaTeX renders inline; the Session panel tracks rounds and exposes "Grade My Answers" / "Next Round" controls.



Left: concept coverage row — each cell is one concept colored by highest difficulty exercised; tick marks delineate dependency depths and the bolder tick marks ***focus_depth***. Right: generation options panel.

Pilot Evaluation (N=2)

Offline harness over the shipped `dataset.json` via Braintrust. No baselines; our system only.

Scorer	Mean	Min	Max
NoRunError	100%	1.00	1.00
FeasibilityAlign	100%	1.00	1.00
OutputNonEmpty	100%	1.00	1.00
FoundationCoverage	100%	1.00	1.00
ProblemPromptShape	100%	1.00	1.00
AnswerLength	100%	1.00	1.00
BodyLengthBand	100%	1.00	1.00
ProxyTokenEfficiency	100%	1.00	1.00
RubricSatisfaction (LLM judge)	91%	0.82	1.00
Latency (band score)	87.5%	0.75	1.00
ItemStructural	85%	0.70	1.00
ItemClarity (LLM judge)	69%	0.38	1.00
MarkdownStructure	46%	0.35	0.57

With $N=2$ these numbers characterize the harness, not a benchmark; CIs are wide and unreported, means are descriptive only.

Post-pilot additions

Three features landed after the eval was frozen — covered by unit tests but not yet in the table:

1. Per-concept Answer Evaluator + Elo mastery loop.
2. Multimodal Sample Evaluator (markdown or exam-clip image).
3. Depth-aware scheduler + concept coverage visualization.

Responsible AI

Raw notes stay in the vault. Cloud LLM and embedding calls are opt-in. The multimodal sample evaluator base64-encodes images in the browser before forwarding — a new exfiltration surface that is surfaced in-session only and deduplicated across evaluator and generator calls. Per-concept verdicts also reduce a known grading-drift failure mode where a single scalar score punishes concepts the learner never even attempted.

Limitations & Future Work

- In-memory sessions; not durable across restarts.
- Groq path is text-only; sample images stripped with a warning.
- MinerU PDF ingest brittle on multi-column layouts and low-quality scans.
- Future: stratified eval, human ratings, durable mastery storage, Groq multimodal, larger labeled trajectory set for the Answer Evaluator.

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